Harrow Creek Water Treatment Plant

1 Design Basis

The Harrow Creek Water Treatment Plant is sized for a design flow of 4,085 m³/h (average day) and a peak flow of 6,656 m³/h on the maximum day, serving a population of 42,680 persons.

Measured over five years, the 95th percentile raw water turbidity is 37.6 NTU. Finished water turbidity shall not exceed 0.23 NTU in 95% of monthly samples.

2 Treatment Process

The rapid mix basin provides a detention time of 53 s at design flow, with a velocity gradient of 922 1/s. Flocculation follows in 4 stages, for a total detention time of 23.5 min at design flow.

There are 2 sedimentation basins, each 48.5 m long and 14.3 m wide, with a side water depth of 4.6 m; the surface loading rate is 0.86 gpm/ft² at design flow.

Filtration uses 8 dual-media filters of 404 ft² each. With one filter out of service, the filtration rate is 4.1 gpm/ft².

Table 1. Raw water pumps

Tag	Capacity (L/s)	TDH (m)	Motor (kW)
P-101A	325.0	29.5	199
P-101B	352.7	19.8	104
P-101C	538.9	21.2	173

3 Chemical Feed

Alum is dosed at 38.8 mg/L on the average day and up to 55.7 mg/L on the maximum day. Chlorine is fed at 3.3 mg/L ahead of the clearwell.

4 Disinfection Options

Two options were compared for primary disinfection.

Table 2. Disinfection options

	Option A: UV	Option B: chlorine contact
Capital cost ($M)	6.35	2.82
Annual energy use (MWh/yr)	771	n/a
Contact time at peak flow (min)	n/a	49